

2 Comparative Suppletion

2.1 Introduction

In many languages, a handful of adjectives form their comparative grade via a root (or base) that is etymologically unrelated to the positive root. It is for this phenomenon that the term *suppletive* (German: *suppletorisch*) was originally coined by Osthoff (1888, 1899). Some examples are given in (28), with roots boldfaced.¹

(28)	<i>POS</i>	<i>CMPR</i>	
a. English	good	bett -er	
b. Latin	bon -us	mel -ior	‘good’
c. Icelandic	gamall	eld -ri	‘old’
d. Georgian	k’argi -i	u- mjob -es-i	‘good’
e. Upper Svan	žɣəd	xo-š-a	‘big’
f. Abkhaz	a- c^wgⁱa	j- ejc^wo -w	‘bad’
g. Basque	on	hobe	‘good’
h. Estonian	hea	pare -m	‘good’

In most such examples, the suppletive comparatives have regular comparative morphology but the root is not supplied by or phonologically derived from the positive base. Pairs of suppletive and nonsuppletive comparatives in (29) illustrate this. (In a small minority of examples, such as English *bad* – *worse*

1. For familiarity, examples will generally be given in the standard orthography or standard transliteration of the languages in question, and in citation forms wherever practical. Thus, even though our interest is in the roots or stems and the comparative and superlative morphology, many examples will contain additional (irrelevant) inflectional material, such as the masculine nominative singular ending *-us* in Latin (28b) and elsewhere.

and the Basque example in (28g), the comparative morpheme is missing, a situation I return to in sections 2.2 and 5.3.1.)

(29)		<i>POS</i>	<i>CMPR</i>	
a. English	good	bett-er		
	long	long-er		
b. Latin	bon-us	mel-ior		‘good’
	dur-us	dur-ior		‘hard’
c. Icelandic	gamall	eld-ri		‘old’
	stór	stær-ri		‘strong’
d. Basque	asko	gehi-ago		‘most’
	zoro	zoro-ago		‘crazy’
e. Votic	üvä	parə-pi		‘good’
	süvä	süve-pi		‘deep’

Adjectives in some languages also have a superlative grade, with the meaning ‘more ADJ than all others’. (Recall from chapter 1 that throughout this work, unless otherwise noted, I will restrict the term *superlative* to the sense of relative superlatives, excluding the “absolute” superlatives such as Italian *bell-issim-a* ‘very beautiful’.) Suppletion also extends to the superlative grade. In the majority of cases, the superlative and comparative forms share a common root, distinct from that of the positive (Ultan 1972, 144), as in (30). I will refer to this as an ABB pattern, to indicate that the shared root (B) in the comparative and superlative is distinct from the root of the positive (A).

(30)	<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>	
a. English	good	bett-er	be-st	
b. English	bad	worse	wor-st	
c. Danish	god	bed-re	bed-st	‘good’
d. Czech	špatn-ý	hor-ší	nej-hor-ší	‘bad’
e. Georgian	k’argi-i	u-m’job-es-i	sa-u-m’job-es-o	‘good’
f. Estonian	hea	pare-m	par-im	‘good’
g. Kildin Saami	šig’	pər’-am	pər’-mus	‘good’
h. Basque	asko	gehi-ago	gehi-en	‘many’
i. Kabardian	kwad, ba	nax	nax-deda	‘many’
j. Cherokee	osda	dajehla	wi-dajehl-āʔi	‘good’

Another suppletive pattern that is rare, but nevertheless attested, is the ABC pattern, in which each grade is built on a distinct root. The Latin, Welsh, Old Irish, and Middle Persian triples meaning ‘good – better – best’ constitute ABC patterns, as shown in (31).

(31)		<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>	
a. Latin	bon -us	mel -ior	opt -imus		‘good’
b. Welsh	da	gwell	gor -au		‘good’
c. Old Irish	maith	ferr	dech		‘good’
d. M. Persian	xōb	weh/wah -īy	pahl -om/ pāš -om		‘good’

I will argue below that ABB and ABC are the only attested suppletive patterns and that a handful of apparently divergent patterns should be reanalyzed. The state of affairs regarding comparative suppletion may then be schematized as in (32). Of five logically possible patterns of root identity and suppletion, only three are attested.

(32)		<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>	
a. Regular	A	A	A	<i>big – bigger – biggest</i>	
b. Suppletive	A	B	B	<i>good – better – best</i>	
c. Doubly suppletive	A	B	C	<i>bonus – melior – optimus</i>	
d. Unattested	A	B	A	<i>*good – better – goodest</i>	
e. Unattested	A	A	B	<i>*good – gooder – best</i>	

This patterning was noted briefly in the closing paragraphs of the only previous study of the morphology of comparison (Ultan 1972). For a 20-language sample, Ultan (1972, 144) notes that “suppletive paradigms in the comparison of adjectives almost always imply formal identity or near-identity of the bases shared by the comparative and superlative vis-à-vis those shared by the positive and equative.” Ultan’s generalization is robustly supported in the larger survey reported here. Indeed, I submit that (a slight reformulation of) Ultan’s generalization is a strong contender for the status of a linguistic universal.² For reasons that will become clear as we proceed, I suggest breaking the generalization into two pieces, to cover (32d) and (32e) separately. We may thus formulate the two-part generalization in (33) and (34), referring to the whole generalization as the CSG.

(33) *The Comparative-Superlative Generalization, part 1 (CSG1)*

If the comparative degree of an adjective is suppletive, then the superlative is also suppletive (i.e., with respect to the positive).

2. Ultan suggested that the generalization represents a strong tendency and not an absolute. In part, this is because the ABC pattern is anomalous as regards his generalization (showing nonidentity between the comparative and superlative); the ABC pattern is consistent with the CSG as formulated below. In addition, Ultan believed ABA patterns to occur. However, the one apparent ABA pattern he noted seems to be misanalyzed (see footnote 11 of chapter 4).

(34) *The Comparative-Superlative Generalization, part 2 (CSG2)*

If the superlative degree of an adjective is suppletive, then the comparative is also suppletive (i.e., with respect to the positive).

The CSG (both parts) ranges over synthetic (i.e., morphological) comparative and superlative grades. Analytic (periphrastic) constructions show different behavior (I return in section 3.4 to why this is so). The immediate goal of this chapter is to propose an explanation of this apparent universal and to consider some consequences of that explanation. The explanation I offer has two key components (both qualified below). First, the representation of the superlative must properly contain (be derived from) that of the comparative. Second, suppletion constitutes a special case of rules of exponence (also called vocabulary insertion), introducing distinct root formatives into specific contexts. Such rules are most properly formulated in terms of a *realizational* theory of morphology (in the sense of Stump 2001), in which morphological rules spell out a derivationally prior complex (i.e., syntactic) structure. Part 1 of the CSG in (33) follows essentially from these two assumptions. An additional assumption, that there is an adjacency condition on context-sensitive rules of exponence, extends the theory to account for part 2 of the CSG.

The remainder of this chapter is organized as follows. In section 2.2, I first sketch the bare bones of the analysis, presenting the leading ideas but leaving details and motivation of assumptions aside. I then turn to a type of poverty-of-the-stimulus argument raised by the suppletion data. The key generalizations concern gaps, and since suppletion is such a marginal phenomenon in any one language, there is no way that the data available to any given learner could be robust enough to distinguish accidental from systematic gaps in his or her language. In section 2.3, I elaborate on this argument for the hand of UG, arguing against an appeal to historical accident.

Once the reader can see where we are headed, I turn in chapter 3 to a discussion of independent evidence for one key assumption, and then I present a refinement of some details of the theory, necessitating many asides (some of which reveal fruitful additional generalizations).

In chapter 4, I lay out in more detail the results of a crosslinguistic investigation of (33). In a relatively comprehensive survey of comparative suppletion encompassing more than a hundred distinct examples (there are some tricky issues in counting, discussed below), only a handful of *prima facie* counterexamples turn up, for which I propose alternative analyses in sections 4.1 and 4.3. If the apparent counterexamples can indeed be explained away, as I sug-

gest, then the ABB and ABC patterns are indeed the only attested ones. No adjective shows an unambiguous ABA pattern—that is, hypothetical **good – better – goodest* or **bonus – melior – bonissimus* in which the comparative alone is suppletive, with the positive and superlative sharing a common root.

2.2 *ABA: Explaining a Gap

The account of the CSG begins by assuming that the representation of the superlative properly contains that of the comparative in all languages. I call this the *Containment Hypothesis*. Note that this embedding is transparent in the overt morphology in many languages, as in the Czech and Georgian examples in (30) and in the nonsuppletive examples in (35).

(35)	POS	CMPR	SPRL	
a. Persian	kam	kam -tar	kam -tar-in	‘little’
b. Cimbrian	šüa	šüan -ar	šüan -ar-ste	‘pretty’
c. Czech	mlad -ý	mlad -ší	nej- mlad -ší	‘young’
d. Hungarian	nagy	nagy -obb	leg- nagy -obb	‘big’
e. Latvian	zil -ais	zil -âk-ais	vis- zil -âk-ais	‘blue’
f. Ubykh	nüs [°] ə	ç’a- nüs [°] ə	a-ç’a- nüs [°] ə	‘pretty’

Further evidence for the Containment Hypothesis (and some qualifications) will be provided in chapter 3. The most straightforward (but not the only) means of representing containment is via a nested structure, in which the superlative is derived from the comparative by the addition of some morpheme. Thus, we have the *Nesting Hypothesis* in (36a) as a special case of the Containment Hypothesis.³ Under certain assumptions, this can be seen as an implementation of the markedness hierarchy positive < comparative < superlative proposed by Greenberg (1966), Canger (1966), and Ultan (1972), a topic to which I will return.⁴ On the Nesting Hypothesis, despite appearances,

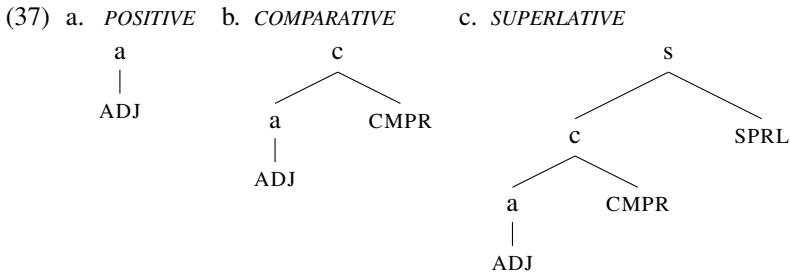
3. Here and throughout, diagrams represent stems to which inflectional morphology may be added. The comparative embeds the positive stem (or base), minus inflectional morphology.

4. The Nesting Hypothesis is also reminiscent of proposals for an implicational hierarchy among a series of functional heads in particular domains, where the inventory of functional heads is subject to crosslinguistic variation, but the presence of a higher head entails the presence of lower heads. See Wurmbrand 2008, 2011c, for a proposal regarding variation in the structure of infinitival complements, and Radkevich 2010 on PP structure.

the representation in (36b) is thus incorrect for languages like English; there must be a “hidden” comparative element even in forms such as *biggest*.⁵

- (36) a. [[[ADJECTIVE] COMPARATIVE] SUPERLATIVE]
 b. *[[ADJECTIVE] SUPERLATIVE]

The structure in (36a) can of course be represented as a tree diagram, as in (37). I use the convention of SMALL CAPS to represent the abstract morphemes, and I put aside questions about the proper labels for nonterminal nodes. Thus, *a*, *c*, *s* are simply mnemonics to refer to specific nodes in the diagrams. Also, as in (36a), I represent the affixes as suffixes, though what is of interest is the hierarchy (constituency), not linear relations (e.g., the superlative exponent is a prefix in Czech).



Consider now the nature of the rules of exponence that insert phonological material at the nodes in (37). The first case to consider is Czech, a language in which the nesting structure in (37) is morphologically transparent, as noted in descriptive grammars of the language (e.g., Janda and Townsend 2000). The relevant forms of two Czech adjectives are given in (38).

- (38) POS CMPR SPRL
- a. **mlad-ý** **mlad-ší** nej-**mlad-ší** ‘young’
- b. **špatn-ý** **hor-ší** nej-**hor-ší** ‘bad’

A fragment of the Czech rules of exponence (the Vocabulary) is given in (39).⁶

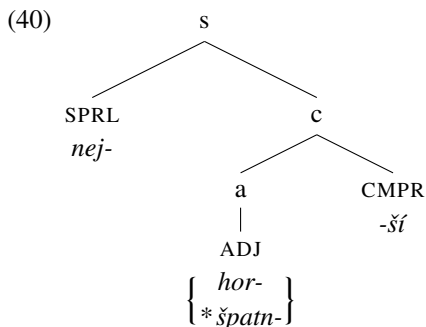
5. Note also that I must reject a POS morpheme in the positive, a morpheme that, although not clearly manifest in overt morphology in any known language, is an important component of some semantic treatments; see Kennedy 2007b. I return to this question in section 4.2.

6. Here and below, the rule fragments presented are only those required for the point under discussion. The Czech Vocabulary includes additional rules introducing other allomorphs of the comparative suffix, notably *-ější/-ejší*, along with contextual restrictions regulating their distribution, and the same holds for comparative and superlative exponents in other languages below. These complexities do not affect the structure of the argument and are left aside for ease of exposition.

- (39) a. SPRL → nej-
 b. CMPR → -ší
 c. YOUNG → mlad-
 d. BAD → hor- / ____] CMPR]
 e. BAD → špatn-

The rules in (39a–c) account for the three forms of the regular adjective *mlad-ý* (additional rules insert inflectional exponents, including the masculine singular nominative -ý in the citation form). The exponents simply realize the corresponding terminal nodes in the structures in (37).

Where a regular adjectival root such as *mlad-* ‘young’ has only a single form, suppletive roots have, by definition, multiple allomorphs. The Czech root meaning ‘bad’ has two: one (*hor-*) introduced by the context-sensitive rule in (39d), and the other (*špatn-*) introduced by the context-free rule in (39e). The logic of elsewhere ordering ensures that the more specific allomorph is chosen whenever possible. Thus, both allomorphs compete for insertion in the comparative structure in (37b), and the allomorph *hor-*, being the most specific form available, wins. In the positive structure (37a), the environment for *hor-* is not met, and thus the elsewhere allomorph *špatn-* wins by default. Now notice what happens in the superlative. By assumption, the representation of the superlative (properly) includes that of the comparative. Therefore, the context for the “comparative” allomorph *hor-* is met, and that allomorph is selected. The rules in (39a–b) operate as before, supplying exponents to the comparative and superlative nodes. The superlative, with the proper exponents, is shown in (40).



This constitutes the first key result: the combination of assuming a nested structure and applying the elsewhere logic to root allomorph selection yields an ABB pattern as an automatic consequence whenever an adjective has a suppletive comparative and nothing further is said. Because the comparative is contained in the superlative, the comparative allomorph (of an adjectival root)

will automatically be compatible with the superlative context as well, and will necessarily block the positive (default) allomorph of that root. Thus, the ABA pattern (a return to the positive root in the superlative) is effectively excluded, yielding the core content of the CSG1.

Now, although the logic just sketched correctly excludes ABA patterns, it nevertheless does allow for patterns beyond ABB. In particular, the ABC pattern is readily describable. The comparative root allomorph will necessarily block the positive in the superlative context, but the comparative allomorph can in turn be blocked by an even more highly specific exponent, as in the allomorphs of the Latin root for 'good'. The Vocabulary fragment in (41) provides the rules of exponence that will derive the ABC pattern for this adjective (in chapter 5, I will revise the somewhat cumbersome formalism for the superlative context in (41a)).

- (41) a. GOOD $\rightarrow$ opt- / ____] CMPR] SPRL]
 b. GOOD $\rightarrow$ mel- / ____] CMPR]
 c. GOOD $\rightarrow$ bon-

 d. SPRL $\rightarrow$ -imus
 e. CMPR $\rightarrow$ -ior

As it happens, one additional clean-up is needed for the Latin examples. The rules in (41) provide the correct root allomorph in each context, but will generate an overt comparative suffix inside the superlative. While this is correct for Czech (and many other languages; see chapter 3), the rules as stated will incorrectly give **opt-ior-imus* in place of *opt-imus* for Latin. To derive the surface forms in which the comparative marker is not visible in the superlative, I posit that the comparative morpheme has a phonologically null allomorph that occurs in the context of the superlative, as in (42). The standard elsewhere logic will ensure that the null allomorph wins out over the regular comparative (43b) in the superlative, just as it selects the correct root exponents in the suppletive cases so far examined.

- (42) CMPR $\rightarrow$ $\emptyset$ / ____] SPRL]

The revision to the context for (41a) to be offered in chapter 5 will in fact avoid the need for (42) for the Latin examples just given, but the issue is a more general one, and a null allomorph has broader application. For example, adding (42) to the English rules of exponence in (43) will correctly generate *bigg-er*, as opposed to **bigg-er-est*.

- (43) a. BIG → big
 b. CMPR → -er
 c. SPRL → -est

The assumptions above serve to effectively derive the CSG1 from the Containment Hypothesis and the assumption that rules of exponence are subject to elsewhere ordering: the ABA pattern is unstatable as a formal pattern. If there are only two distinct listed root forms in a language's Vocabulary, no ordering of the rules introducing these roots will lead to an ABA pattern. But the attested patterns, ABB and ABC, are readily derived, with ABB in some sense being the unmarked case of comparative suppletion, as against the ABC pattern, which requires an additional rule.

Now, there is a loophole, in the sense that it is in principle possible to generate a surface ABA pattern by appeal to accidental homophony. The grammatical pattern would necessarily be an ABC pattern, with three formally distinct root allomorphs, but where (the analogue of) (41a) would introduce the same phonological matrix as (41c), sandwiching the comparative allomorph between them. I suggest that the grammar per se does not exclude this possibility, but I invoke the well-worn idea that there is a general antihomophony bias in acquisition, of which (44) is a special case.

(44) *Antihomophony*

Learners avoid positing a contextual allomorph of a morpheme μ that is homophonous with the default exponent of μ .

If one thinks of rules of exponence as overwriting or rewriting rules, the essence of (44) is that learners will posit no rule of exponence of the form $X \rightarrow X$, which effects no change but serves only to block some more general rule. Irregular forms that appear to have no change from the base, such as the English irregular plural *sheep* (singular *sheep*) or past tense *hit* must thus involve a zero affix (contra Anderson 1992). The proposal in (44) closes the loophole that would otherwise allow for false ABA patterns, arising as ABC patterns in disguise, despite the grammar universally excluding true ABA patterns. This leaves some wiggle room; one could consider the homophony strategy as a retreat of last resort with the implication perhaps that observationally, the CSG1 should emerge as a trend, rather than an absolute (cf. Pertsova 2007). Accidents do happen, after all. I will nevertheless resist this, and at least as a working strategy I will pursue the stronger claim that (44) is never violated and thus that apparent ABA patterns must find alternative explanations. I return to this point in chapter 3 and section 4.2.

Up to the loophole just mentioned, the preceding paragraphs have served to derive the CSG1 as a straightforward consequence of two main assumptions: the Containment Hypothesis and elsewhere ordering. Another component of morphological theories is the question of locality. Various authors across a variety of frameworks have argued for an adjacency (or contiguity) condition on allomorphy, such that the trigger for suppletive allomorphy must be adjacent to the root that undergoes allomorphy (see Siegel 1978; Allen 1979; Embick 2003, 2010; also see Hay 2000 for a brief review of the earlier proposals). Adopting this assumption will serve to extend the account above to the CSG2, excluding the unattested *AAB pattern **good – gooder – best*, as in (32e). Under the Containment (specifically, the Nesting) Hypothesis, an attempt to derive an AAB pattern via the vocabulary items in (45) will fail, as the context in (45a) falls afoul of the adjacency condition.

- (45) a. GOOD $\rightarrow$ be(tt)- / ____] SPRL]
 b. GOOD $\rightarrow$ good

Of course, care must be taken in formalizing this such that whatever device is used to condition the superlative root allomorph in the ABC cases (see (41a)) must be unavailable for putative, but unattested, *AAB cases such as (45). I postpone further discussion of this point until section 5.3, concentrating on the *ABA cases until then.⁷

Consider now what would happen if the Nesting (or Containment) Hypothesis were not adopted. Standard descriptions of English (including the majority of formal treatments of the superlative, such as Szabolcsi 1986, Heim 2000, and Hackl 2009) posit two morphemes, the comparative and the superlative, both of which attach to the adjective directly, as in (46).

- (46) a. [[ADJECTIVE] COMPARATIVE/-ER]
 b. [[ADJECTIVE] SUPERLATIVE/-EST]

A variation of this description posits a single *degree* head (DEG) as in (47), of which both the comparative and the superlative are possible values.

7. Note further that the rule introducing the zero comparative in (42) is formulated as a rule of exponence. On the assumption that vocabulary insertion proceeds cyclically from the root outward, (42) thus automatically follows vocabulary insertion at the root. Rule (42) must be prevented from deleting or pruning the comparative node prior to root insertion; otherwise, it will render the root and the superlative adjacent, incorrectly feeding rules like (45a) and bleeding those like (39d). The approach to impoverishment in Trommer 1999 treats all impoverishment rules as special cases of zero-vocabulary insertion, which would ensure the correct ordering in the cases at hand. See also Embick 2010 for discussion of pruning and adjacency.

(47) [[ADJECTIVE] DEGREE]

Theories starting from these assumptions will be able to describe the attested patterns of root suppletion (although something more needs to be said in order to insert both comparative and superlative morphology in languages with overt nesting, like Czech). Under (47), the ABB pattern among the roots in Czech could be described as in (48) (compare with (39) above), by making the degree head (common to both comparative and superlative) the context for the allomorphy.

- (48) a. BAD → hor- / ____] DEG]
 b. BAD → špatn-

Although reference to DEG may look like a convenient means of representing the ABB patterns, it appears to fail for languages that have more degree affixes than just comparative and superlative.⁸ Ultan (1972) notes that the common pattern of shared suppletive roots groups comparative and superlative together, to the exclusion of the equative degree, even where this can be affixal, as in Welsh (49).

- (49) POS CMPR SPRL EQUATIVE
 drwg **gwaeth** **gwaeth-a** cyn-ddrwg ‘bad’

The same point can be made with reference to the intensified or absolute superlative degree, marked in Czech by the prefix *pře-* (as in (50a–b)) or by other intensifiers such as the suffix *-ink-* (50c). In both these cases, positing a suppletive allomorph in the context ____] DEGREE] would incorrectly overgeneralize that allomorph to the equative and intensified or absolute superlative forms, respectively.⁹

8. My thanks to Jessica Rett for this observation.

9. I thank Ivona Kučerová and Pavel Caha (personal communication) for discussion of the Czech forms. Both note that *pře-dobr-ý* is somewhat stilted or “bookish,” but is clearly the form that fills the slot indicated in (50a), as opposed to a form built on the root *lep-*.

The point here is that suppletion in the comparative automatically extends to the superlative, unless there is further suppletion (an ABC pattern), but the suppletion in the comparative does not automatically extend to the equative. Nothing prevents the comparative allomorph from being specified to occur in the equative context as well. Thus, Welsh *bach* ‘small’ has the suppletive allomorph *llai* ‘smaller’, which occurs also in both the superlative and the equative; Welsh *da* ‘good’ has four distinct roots, one for each grade (including the positive).

- | | | | | | |
|------|----------------|---------------|--------------------|---------------------|---------|
| (50) | <i>POS</i> | <i>CMPR</i> | <i>SPRL</i> | <i>A.SPRL/INTNS</i> | |
| a. | dobr-ý | lep-ší | nej- lep-ší | pře- dobr-ý | ‘good’ |
| b. | špatn-ý | hor-ší | nej- hor-ší | pře- špatn-ý | ‘bad’ |
| c. | mal-ý | men-ší | nej- men-ší | mal-ink-ý | ‘small’ |

Even if we restrict the discussion to languages without these extra degrees, the ABC patterns establish that reference to degree alone is insufficient and that statements of contextual allomorphy must be able to distinguish between comparative and superlative. Thus, under (46) or (47), Latin would require the following treatment:

- (51) a. GOOD → opt- / ____] SPRL]
 b. GOOD → mel- / ____] CMPR]
 c. GOOD → bon-

And therein lies the rub. Consider a pattern exactly like Latin, but without the rule in (51a). In a theory with the Containment Hypothesis, elsewhere ordering ensures that the allomorph in (51b) will automatically extend to the superlative environment, yielding the ABB pattern. But in a theory lacking the Containment Hypothesis, no such extension is guaranteed. The comparative environment is just that—the comparative—and the basic pattern is thus the unattested *ABA (**bonus* – *melior* – *bonissimus*). Put differently, without the Containment Hypothesis, nothing precludes direct reference to the comparative context alone, and it is this possibility that prevents the theory from deriving (hence explaining) the CSG. At the risk of redundancy: the argument here is not that the theories differ in their abilities to describe the attested patterns; what is at stake is whether the unattested *ABA pattern is excluded by the theory for principled reasons.

This concludes the presentation of the core analysis. There are numerous refinements and clarifications to be presented, and it is these—as well as more careful discussion of the empirical basis for the claims, and apparent counterexamples—that will occupy the next chapters. One obvious question is why the Containment Hypothesis would hold, and relatedly, how it might be formally cast within UG. I will postpone discussion of this until chapter 7, offering only a rather incomplete speculation even there. As noted in chapter 1, what I will suggest is that UG imposes limits on the complexity of functional morphemes: the combination of pieces needed to compose a superlative (namely, the comparative and something yielding ‘than all (others)’) is simply too big to fit into a single functional head. The Containment Hypothesis is thus not formally a part of UG; rather, it is a corollary of the complexity of the superlative’s meaning. Before addressing the details of the morphological theory, and the empirical minutiae, I wish to step back to consider (and reject)

an alternative approach to the CSG that would treat it as a historical accident, seeking an externalist explanation rather than the internalist mechanism of UG.

2.3 Universal Grammar versus the European Sprachbund

In the preceding section, I have sketched a preliminary account of the CSG. The account derives this generalization from the Containment Hypothesis and from the assumption that suppletion is to be modeled as contextual allomorphy, implemented by rules of exponence (vocabulary insertion), subject to elsewhere ordering. Arguments from gaps (of the sort characterized by the CSG) rest implicitly on the premise that the missing pattern is unlikely to be merely an accidental gap. This is essentially a statistical argument, even if it is not formalized as such, and this raises two points within the current context.

2.3.1 The Insignificance of the Evidence

The first point is a newish twist on the familiar poverty-of-the-stimulus type of argument for UG. If the CSG is a valid generalization, as I contend here, then it is not something that can be learned from the data available to a normally developing child. Since suppletion is such a marginal phenomenon internal to any one language, with so few relevant forms (if any at all), the primary linguistic data to which any one child is exposed are far too sparse to warrant any conclusions about impossible suppletive patterns. In other words, the absence of the ABA pattern in any given language is in and of itself insignificant. The generalization is only significant, and thus worthy of attention, in its crosslinguistic context.

This point can be appreciated with reference to English, which has a handful of suppletive comparatives (*better*, *worse*, *more*, *less*), all of which participate in ABB patterns. The ABA pattern is indeed absent from English, but with only four triples (two of which have further irregularities), it would seem rash to infer that this is a significant absence, indicative of a general property of UG. Indeed, the ABC pattern is just as absent from English as the ABA pattern is, yet it would be simply false to conclude that the ABC pattern is disallowed by UG, as it is indeed attested in other languages (Welsh and Latin examples were given above). Other languages show a similar situation, many with but a single element entering into suppletion (and the vast majority having of course none whatsoever).

The explanation of the contingent fact that Modern English *good* compares on an ABB pattern *good* – *better* – *best*, and not some other pattern, certainly

lies squarely in the history of the language: this pattern is inherited from previous stages of English (ultimately from the earliest common Germanic ancestor). If the ancestor of Modern English had had a different pattern (say, along the lines of *good – better – finest*), then that pattern would presumably have been the one Modern English inherited. The existence of ABB and ABC patterns demonstrates that children are equipped to acquire those patterns on the basis of evidence in the input, and the account of what any one child acquires needs no appeal to UG beyond the general ability to learn these patterns from the input. Where UG is invoked instead is at the level of the broad, crosslinguistic generalization. UG explains (in the manner laid out in the previous section) why, from among the many suppletive patterns that have arisen in a variety of languages, no genuine ABA or AAB patterns are to be found. In an important sense, then, UG places bounds on possible language change (cf. Kiparsky 2008): no language can undergo a change that yields a genuine ABA or AAB pattern, since no learner could posit a grammar that would accommodate such a pattern.

In sum, the logic here is that the CSG cannot be learned from the data. Therefore, either it must be spurious, holding true only accidentally (I argue against this chapter 4), or it must be attributed to some more general constraint imposed on possible grammars that a learner may posit. In principle, this yields two options: either an inherent grammatical constraint (UG), or a language-external consideration, such as some general principle of cognition. So far as I can see, there are no current candidates for a general cognitive or functional principle that would exclude the ABA or AAB patterns. For example, the markedness hierarchy *positive < comparative < superlative* is discussed in the functionalist literature by Heine (1997, 124–126); yet rather than providing independent evidence for some language-external consideration that will derive the hierarchy, Heine approaches the matter from the other direction, arguing that the evidence for this hierarchy is (thus far) purely linguistic, and suggesting therefore that some (as yet unknown) functional pressure must exist. Absent a cognitive account, the best alternative is that some a fortiori constraint on mental grammars derives the CSG. I have offered the Containment Hypothesis (in tandem with elsewhere ordering) as the relevant constraint, as a property of UG.

2.3.2 Counting Cognates

This brings us to the second point about taking the evidence for the CSG to be significant only in the broader, crosslinguistic context, and it is a more narrowly methodological one. While there are numerous examples of suppletion in the material assembled here, and while the handful of appar-

ent counterexamples seem likely to fall to an alternative account (see chapter 4), there is nevertheless a very restricted distribution (in both geographic and genetic terms) to the component phenomena of the CSG (suppletion in adjectival gradation and morphological comparatives and superlatives). Outside of the Greater European Sprachbund (Indo-European, Finno-Ugric, Kartvelian, Northwest Caucasian, and Basque), the CSG is trivially true. With the notable exception of the Cherokee example in (30j), the languages that supply the key evidence represent a contiguous, albeit large, geographic area. This raises the concern that the CSG, even if descriptively valid over hundreds of examples from numerous languages, could at its core be an effect of shared vocabulary, to be explained by a mixture of common inheritance and borrowing (or calquing) under contact, rather than through an appeal to a universal structural characteristic.

I suggest that this worry can be mitigated by looking more closely at the individual examples and being more careful about what we count in constructing the quantitative argument that the gap is not accidental. First, we must give up counting languages. In large geographically and genetically diverse samples, counting languages (as commonly understood) may be reasonable at getting a rough measure of the distribution of some phenomenon (see Bickel 2008 and references therein for relevant discussion). But in the case at hand, language counting faces intractable problems. Most obvious among these is a definitional problem: are Norwegian and Swedish (or Serbian and Croatian) one language, or two, or more? How many varieties of English are there? Even if we settle on some (essentially arbitrary) measure, the fact remains that many of the attested suppletive paradigms are not independent of one another. Consider, for example, the following Germanic paradigms for ‘good’:

(52)

	<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>
a. English	good	bett-er	be-st
b. German	gut	bess-er	(am) be-st-en
c. Gothic	gop-s	bat-iz-a	bat-is-t-s
d. Afrikaans	goed	bet-er	be-ste
e. Swedish	god	bätt-re	bä-st
f. Cimbrian	guat	pez-ar	pez-ar-ste

Almost all Germanic languages, spanning the globe from Norwegian to Afrikaans, from the oldest attested (Gothic) to the present, share a paradigm whose pieces are cognate with English *good – better – best*. Undoubtedly, this pattern arose exactly once, at some point in the dawn of Germanic (if not before), and the explanation of all of the attested triples is that they have been

inherited from a common source. In weighing the evidence for the CSG, then, these cannot be counted as independent data points; rather, they must collectively be counted as just one. To abstract away from the surface variation, I will use full capitals in representing the roots in the cognate sets. (52) is thus a single cognate set: GOOD – BE(TT) – BE(TT).

This same consideration must apply in evaluating the other examples brought to bear on the evidence for the CSG. In light of the close genetic and areal affinities in the data pool, it is clear we are not drawing from a random sample; instead, we must aim for a comprehensive survey of the individual attested data points, where each is a distinct cognate triple (positive – comparative – superlative). As illustrated with Germanic, whenever there is a common cognate triple, shared vocabulary is the most ready explanation. It is only where the triples are distinct that there must have been some linguistic change, and it is only in the changes that common inheritance cannot provide the entire account.

Pursuing this further, I take it that two cognate sets are distinct from one another if (and only if) they differ in at least one of their component roots, though they need not differ in all. Thus, while (52) represents the majority of Germanic, there are two or three additional cognate sets for Germanic ‘good’, illustrated in (53).

(53)	POS	CMPR	SPRL
a. Swedish	bra	bätt-re	bä-st
b. Giazza Cimbrian	guot	beg-ur	beg-ur-ste
c. Old English	gôd	sêl-(ra)	sêl-ost

Swedish (53a) (also Norwegian) overlaps with (52) in the comparative and superlative, but differs in the positive root; thus, we have BRA – BE(TT) – BE(TT). Another triple, in (53b), is GOOD – WAEH – WAEH from the Cimbrian German variety spoken in Giazza (Italy), which has retained the positive root (cognate to) GOOD, but forms comparative and superlative grades on a root descended from Middle High German *wäh(e)* ‘artful, fine, dainty, good, etc.’ (Schweizer 2008, 397; *DWB*, vol. 27). Old English *gôd* ‘good’ had two comparatives and superlatives, with the forms in (53c) attested alongside a GOOD – BE(TT) – BE(TT) triple (Sievers 1882, 108; Bosworth and Toller 1898, 858). Although these triples overlap in part with GOOD – BE(TT) – BE(TT), each is counted as a distinct triple, as each represents a separate innovation and thus, in a sense, a distinct chance for an ABA pattern to have arisen.

Approaching the matter in this way, I count a total of somewhere just over one hundred distinct suppletive triples; see chapter 4. (Providing an exact

figure still requires some arbitrary choices, as in cases where the relation between positive and comparative forms is not one:one.) Of these triples, a handful pose potential problems for the CSG. One is the Basque word for ‘good’, which shows a suppletive comparative, but a doublet in the superlative: *on – hobe – hobe-ren/on-en*. Others are apparent ABA patterns in the ‘many – more – most’ triples in Karelian, Armenian, and Bulgarian/Macedonian. I discuss these examples in more detail in chapter 4, offering tentative alternative explanations for each case. The remainder are mostly ABB patterns, with a few ABC cases like Latin *bonus – melior – optimus*. No putative AAB cases are attested in the data.

In sum, despite the areal and genetic limitations, there is sufficient variation within the data to be reasonably confident that the *ABA and *AAB gaps are not merely the accidental result of shared vocabulary. They are real and systematic gaps, in need of an explanation.¹⁰

2.3.3 (In)stability and Change

A further argument against attributing the absence of an ABA pattern to shared vocabulary, whether through inheritance or borrowing, is that the vocabulary itself is quite typically not shared across languages, even where the overall pattern is. This can be seen by perusing the cognate lists in table 4.1. While there is certainly an areal effect of having suppletive patterns for core adjectives (‘good, bad, big, many, ...’), there is no evidence to support the claim that the patterns arise from borrowing of adjectives (as lexical items) across languages. Similarly, it is not the functional vocabulary as such (the morphological exponents of comparative and superlative) that is borrowed among languages, leading to the areal diffusion of the nesting pattern.¹¹ Languages

10. A rather naive application of basic statistical tests here yields robust significance, even with the apparent problematic patterns included. But what is really driving this result is the overwhelming number of ABB patterns as against anything else. Within a given language, it is plausible that the ABB pattern for one root influences the abstract pattern for others; thus, without knowing the weight of this effect, applying statistical measures seems at best premature. Relatedly, assuming equal proportionality as the null hypothesis seems highly implausible, but I see no clear way of establishing a baseline chance distribution of patterns.

11. That is not to say that there is no borrowing in this domain. The particles that make periphrastic comparatives (and superlatives) are borrowed, in some cases creating new comparative constructions in the target languages. Some languages of South and Central America have borrowed Spanish *mas* or Portuguese *mais* to form periphrastic comparatives (Tarascan, Chamoreau 2003; Paumari, Chapman and Derbyshire 1991), while indigenous languages of Russia have borrowed Russian *bole* and/or *samyj*. In

with transparent nesting vary widely in the resources they draw on to derive the nested structure (see the next chapter).

Even within a single family, there is significant variation in the actual lexical items (exponents) that make up a given suppletive pattern. Where Germanic shows a remarkable stability in the GOOD – BE(TT) – BE(TT) pattern, not all patterns are nearly as stable. The Slavic ‘good’ patterns show remarkable volatility. In the earliest documented language, Old Church Slavonic (OCS), the form for ‘good’ had a regular comparative, as well as additional suppletive comparative options.¹²

- (54)
- | | | |
|-----|--------------|----------------|
| | <i>POS</i> | <i>CMPR</i> |
| a. | dobrŭ | dobr-ěi |
| b. | — | luč-ŭi |
| c. | — | unj-ŭi |
| (d. | — | sul-ŭi) |

Bulgarian and Macedonian alone among the daughter languages preserve a regular comparative for ‘good’. Exclusively suppletive patterns have arisen in all the other languages, most resolving to a single option, rather than the one:many situation attested in OCS.¹³

a few cases, morphological comparatives and superlatives are borrowed, but in borrowings from Sanskrit and Persian into Hindi (McGregor 1972) and from Tajik into Yagnobi (Khromov 1972), for example, the borrowed morphology is only used with borrowed adjectives. More common is borrowing of forms that constitute suppletive comparatives in the donor language, but lose their comparative morphosyntax in the recipient language, serving as intensifiers, in some cases subject to further comparison or superlative formation. Examples of this sort include the borrowing of Persian *beh-tar* ‘better’ into Hindi and Turkish, and (ultimately) the borrowing of Latin *optimus* into English, where the adjective *optimal* has a meaning like ‘best’ but is not morphosyntactically a superlative form. See section 4.1.

12. The status of the superlative here is somewhat unclear, but orthogonal to the immediate point. Most sources note that the superlative is formed by adding *naj-* to the comparative (e.g., Vondrák 1900, 161) and none note any irregularities in the formation of *naj-* derivatives. On the other hand, Diels (1963, 201n3) claims that OCS had no means of forming a superlative, stating that the *naj-* prefix is very rarely found, and when it is, has in the first instance essentially no superlative meaning. Evidently, the prefix became the standard means of forming morphological superlatives in all the daughter languages, so this quibble is moot here.

13. Superlatives for (55c,f) were not given in the sources from which these were taken, but the sources indicate superlatives formed by prefixing *naj-* to the comparative.

(55)	POS	CMPR	SPRL
a. Bulgarian	dobər	po- dobər	naj- dobər
b. Czech	dobr-ý	lep-ší	nej- lep-ší
c. Sorbian (some varieties)	dobr-y	redl-iši	
d. Serbian	dobar	bol -ji	naj- bol -ji
e. Ukrainian	dobr-yj	krašč -yj	naj- krašč -yj
f. Ukrainian	harn-yj	krašč -yj	
g. Russian	xoroš -ij	luč -še	(nai- luč -š-ij)

The consistency of the Slavic patterning (all suppletive patterns are ABB) is not the result of shared vocabulary, since the vocabulary items (roots) themselves are not shared. None of the suppletive patterns in (55) reflects roots in both the positive and the comparative from the OCS patterns in (54). The Ukrainian and Czech comparative roots were not (so far as available sources indicate) comparatives of ‘good’ in OCS, and *bol*-ji (55d) was one of two suppletive comparatives for ‘bigger’ in OCS (Vondrák 1900, 160; Lunt 1959, 66). Russian retains an OCS comparative root, but the positive root *dobr*- has been usurped by *xoroš*-. What the languages have in common is not the specific vocabulary, but the more abstract ABB pattern (Mel’čuk 2006, 456).

Similarly, a common ABB pattern for ‘small – smaller – smallest’ is found in a variety of Indo-European languages, with a shared comparative root *min*- but variation in the positive roots.¹⁴

(56)	POS	CMPR	SPRL
a. Latin	parv -us	min -or	min -imus
b. OCS	mal -ŭ	mŋj -i	
c. Czech	mal -ý	men -ši	nej- men -ši
d. Gothic	leit -il-s	minn -iza	minn -ists
e. Danish	lille / små	mind -re	mind -st
f. Ancient Greek	mīkr -ós	meí -on	meĩ -st-os

On the basis of such evidence, one might contend, then, that it is not vocabulary that is shared (by borrowing and/or inheritance) but something more abstract, such as the ABB pattern. The account offered here explains why

14. That the Greek forms belong in this paradigm, with the comparative cognate to those in *min*-, but the positive root unrelated, is asserted on the basis of Frisk 1960–1970. The Slavic root *mal*- is related to Germanic *small*, and not to the comparative, according to Vasmer 1953–1958.

that pattern is shared: it is precisely the pattern that arises by default, whenever the comparative is suppletive, if (i) the Containment Hypothesis holds, and (ii) rules of exponence are subject to elsewhere ordering. Simply declaring that abstract patterns may be inherited or borrowed does not provide any clear alternative; it merely restates the description that the UG-based account aims to explain. Moreover, there is clear evidence that the overall patterning is susceptible to change, inasmuch as suppletive patterns rise and fall.

The Slavic examples for ‘good’ illustrate this point. OCS had a one:many pattern in which the suppletive comparatives formed ABB doublets alongside a regular AAA pattern (54a). Yet none of the daughter languages retains the OCS pattern. In this case, simply saying that abstract patterns are retained strikingly fails to explain the facts.¹⁵

Similarly, adjectives that show a suppletive pattern in one language may undergo regularization in daughter languages (ABB → AAA), sometimes giving rise to doublets in the daughter languages, the opposite of the change in Slavic ‘good’ paradigms. Thus, compare Swedish *god* – *god-are* – *god-ast*, a regular triple for ‘good’ in the sense of ‘pleasant-tasting’, or colloquial English *bad* – *badder* – *baddest* (the only morphological comparative form available for *bad* in the sense of ‘cool, hip’; see section 6.4). In Homeric Greek, *agathos* ‘good’ formed only suppletive comparatives, but the adjective regularized in Old Testament Greek (comparative *agathō-ter-os*, superlative *agathō-tat-os*; Liddell and Scott 1996).

In the other direction, innovative suppletive patterns arise where none existed before (AAA → ABB). Swedish *bra* (in (53a)) is one such example (the *Svenska Akademiens Ordbok* gives seventeenth-century *braf* – *brafv-are* – *braw-ast-e*). Another may be OCS *mŕnogŭ* ‘many’, which is given as undergoing regular comparison (*množai* ‘more’; Diels 1963, 201n2; Lunt 1959, 66), but which forms suppletive patterns in the daughter languages (although Krause and Slocum (2002–2004) give a suppletive comparative for this even in OCS). Another innovated suppletive paradigm is the Old Icelandic paradigm for ‘old’. This adjective is regular (more accurately, nonsupplementing) throughout Germanic, but the positive root underwent lexical replacement in Old Icelandic, yielding a new ABB paradigm (retained in the daughter languages).

15. The most extensive cases of one:many relations in suppletive comparatives are found in OCS and Ancient Greek, both written languages representing an area with multiple dialects. There is thus a specter of artificiality about this relationship, though one that does not affect the point made here, namely, that there is variation within Slavic.

(57)		<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>
a. Gothic		alp -eis	alp -iz-a	
b. Old English		eald	ield -ra	ield -est
c. Old Icelandic		gamall	ell -re	ell -ztr

In sum, it is not the case that suppletive patterns as such are immutable. The ABB pattern (and the absence of an ABA pattern) holds a special place in the array of reconstructible changes, and the theory here explains why this is so. By hypothesis, the superlative is always derived from the comparative. Hence, if a suppletive pattern arises—for example, by replacement of the positive root (as in (57c))—then the superlative will not follow suit, but will instead remain tied to the comparative. Likewise, if the comparative changes, then the superlative will change in tandem, tied as it is to the comparative. This holds whether the relevant change is the resolution of doublets (as in (55)), the innovation of a new comparative root (as possibly in Slavic words for ‘many’, mentioned above (57)), or the change by which *little* in Old English was compared by *less* ((58b); cf. (56)).

(58)		<i>POS</i>	<i>CMPR</i>	<i>SPRL</i>
a. Gothic		leit -il-s	minn -iza	minn -ists
b. Old English		lýt -el	læss -a	læs -(as)t

The only permitted exception to the default suppletive pattern of ABB is the possibility for a marked ABC pattern to arise, as in Latin, where three separate roots are listed. ABA does not arise, not because the vocabulary of suppletive patterns is generally inherited, nor because the abstract suppletive patterns are passed down intact from mother to daughter languages as such, but because UG excludes a means to derive a pattern with two roots organized in such a way that the comparative is the odd one out, a single root being shared between the positive and the superlative.

2.4 Summary

In this chapter, I have argued that there is an intimate relationship between comparatives and superlatives and that this relationship is manifested as proper containment of the former in the representation of the latter. There is no (relative) superlative morpheme that attaches directly to adjectives, despite appearances to the contrary in languages like English. The primary evidence for this conclusion, as discussed in this chapter, is the CSG, the relationship between root suppletion in comparatives and root suppletion in superlatives, discussed in the abstract in section 2.2 (a more careful discussion of the data,

and apparent counterexamples, will follow in chapter 4). Because the primary evidence for the generalization comes from a striking gap, and because the strikingness of the gap is apparent only in large, crosslinguistic comparison, the results point to the hand of UG in accounting for this state of affairs. I will return in chapter 7 to a tentative proposal about why this might be the case, but first, I turn in the next chapter to further evidence for the Containment Hypothesis, and to related considerations that arise in that discussion.